EXP.NO: 5 Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

● The user fills out a web form with their name and email.

● The form will use both GET and POST methods for data submission.

2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. Create an HTML Form:

o Design an HTML form with two input fields: one for the name and one for the email.

o Create two buttons to allow the user to submit the form using GET or POST methods.

o Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).

2. Create the Servlet to Handle GET and POST Requests:

o Write a Java Servlet class that will handle both GET and POST requests.

o The servlet will have two methods:

▪ doGet() to handle the GET requests.

▪ doPost() to handle the POST requests.

o In both methods, retrieve the form data using request.getParameter().

o In the doGet() method, process and display the data in the URL.

o In the doPost() method, process and display the data in the HTTP request body.

3. Handle GET Requests in the Servlet:

o In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.

o Output the data as a response, including the name and email in the URL (visible to the user).

4. Handle POST Requests in the Servlet:

o In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.

o Output the data as a response, displaying the name and email without exposing them in the URL (more secure).

5. Return the Output:

o The Servlet should return an HTML page displaying the name and email entered by the user.

o For the GET method, the output should be visible in the URL.

o For the POST method, the output should be visible in the response body, not the URL.

6. Deploy and Test the Servlet:

o Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).

o Test the form using GET and POST methods to see the difference in data submission.

7. Observe and Compare GET and POST Results:

o Test the GET method: The form data is sent in the URL.

o Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

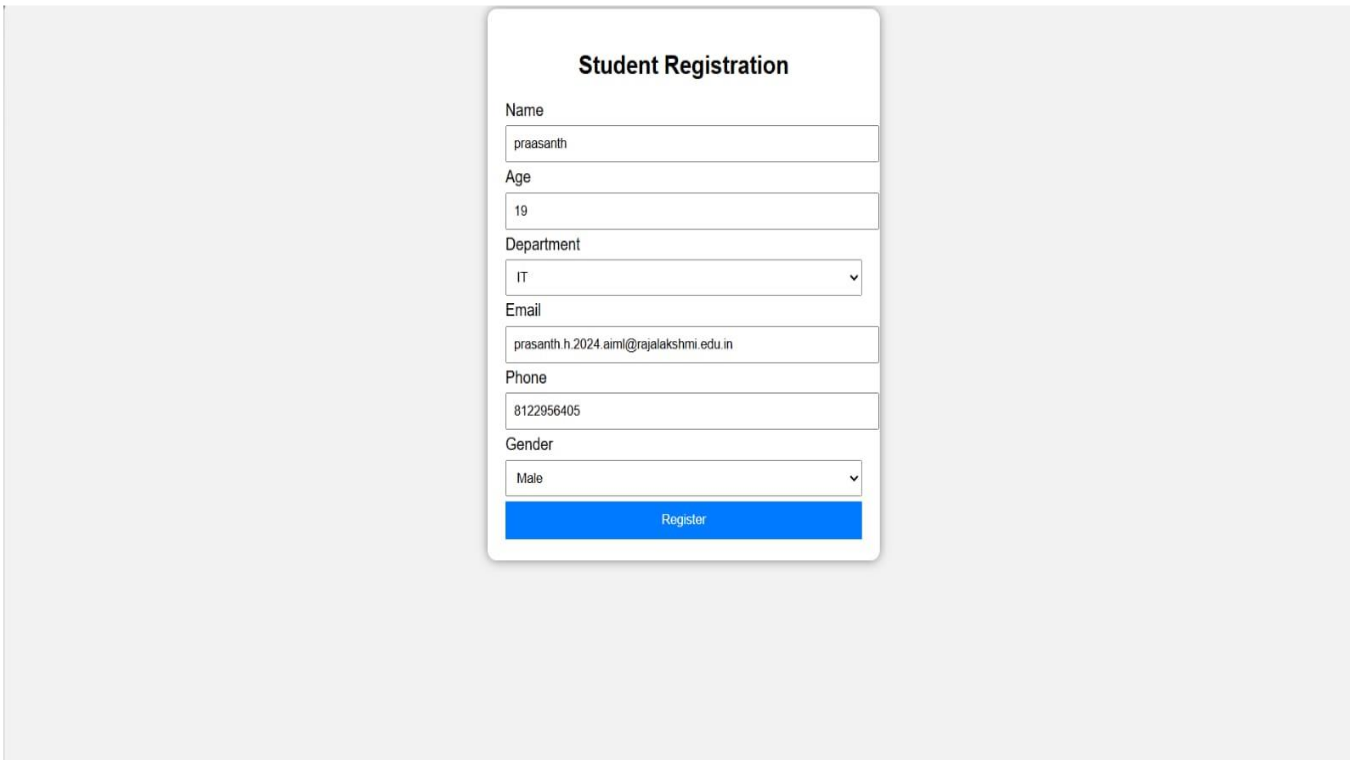
Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Student Form</title>
</head>

<body>

<h2>Student Registration</h2>

<form action="StudentServlet" method="post">

Name: <input type="text" name="name"><br><br>

Age: <input type="number" name="age"><br><br>

Course: <input type="text" name="course"><br><br>

<input type="submit" value="Submit">

</form>
```

```
</body>

</html>

package com.demo;

import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.*;

@WebServlet("/StudentServlet")

public class StudentServlet extends HttpServlet {

protected void doPost(HttpServletRequest request, HttpServletResponse response)

throws ServletException, IOException {

response.setContentType("text/html");

PrintWriter out = response.getWriter();

String name = request.getParameter("name");

String age = request.getParameter("age");

String course = request.getParameter("course");

out.println("<h2>Student Details</h2>");
```

```
out.println("Name: " + name + "<br>");

out.println("Age: " + age + "<br>");

out.println("Course: " + course);


}

}
```

Output:



RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.